

EC-333 Microprocessor and Interfacing Techniques

Experiment # 8

Dynamic Real-time System

1. Objective/s:

- Use change notification interrupts to detect push button presses and releases.
- Eliminate multiple button operations using a timer interrupt to implement a non-blocking delay.
- Implement the push-on / push off operation using a software-based finite state machine
- Lab Task

2. Switch Debouncing Effect

Electromechanical devices, such as switches and relays that are commonly used for microprocessor inputs, are mass-spring devices that will result in contact bounce when opening and closing. Combinational logic circuits will see the multiple contact closures and openings as multiple events unless signal conditioning circuits are used. Usually some type of delay after the initial contact operation is implemented so subsequent operations are ignored for a period of time. The delay can be implemented using an analog low-pass circuit with a resistor and capacitor in addition to electronic devices. Such circuits are costly in terms of circuit board real-estate. As an alternative, there are specialty electronic circuits as well as FPGA designs that can be used to remove contact bounce.

High speed modern microprocessors are also subject to contact bounce issues. Contact bounce issues can be mitigated using simple time delays, such as the polling software delay used in previous lab; however, this delay implementation is a blocking procedure that inhibits the processor from executing code for another task during the delay period. Delays implemented with interrupts are used to notify the processor to execute code for another application task and to notify again when the delay period has expired. This implementation of a delay is non-blocking.

3. Using PIC32 Interrupts with Timer Peripheral

The following steps must be taken in the application code for proper *interrupt exception* initialization & usage.

Step 1. #include Standard Headers

The application must include header files **xc.h** and **attribs.h** as shown below:

```
1#include <xc.h>
2#include <sys/attribs.h>
```

The **<sys/attribs.h>** header file provides macros intended to simplify the application of attributes to interrupt functions:

- **_ISR(V, IPL)**
- **_ISR_AT_VECTOR(V, IPL)**

There are also **Vector Number** macros defined in the processor header files (included via **<xc.h>**)

Step 2. Provide Interrupt Service Routine (ISR)

An interrupt handler function is different to an ordinary function in that it handles the context save and restore to ensure that upon return from interrupt, the program context is maintained. A different code sequence is used to return from these functions as well.

Interrupt Service Routine Definition using the "_ISR()" Macro

The following code example shows how to use the **_ISR()** macro to define an interrupt function:

```
1void __ISR(_INT_VECTOR_NUMBER, IPLx[SRS|SOFT|AUTO]) isrName(void)
2{
3    /* Clear the cause of the interrupt (if required) */
4    /* Clear the interrupt flag */
5    /* ISR-specific processing */
6}
```

The various parameters will now be more fully described.

_INT_VECTOR_NUMBER

- Unique Vector Number Identifier, as shown in Table 7-1 in the device data sheet:

TABLE 7-1: INTERRUPT IRQ, VECTOR AND BIT LOCATION

Interrupt Source ⁽¹⁾	IRQ Number	Vector Number	Interrupt Bit Location			
			Flag	Enable	Priority	Sub-Priority
Highest Natural Order Priority						
CT – Core Timer Interrupt	0	0	IFS0<0>	IEC0<0>	IPC0<4:2>	IPC0<1:0>
CS0 – Core Software Interrupt 0	1	1	IFS0<1>	IEC0<1>	IPC0<12:10>	IPC0<9:8>
CS1 – Core Software Interrupt 1	2	2	IFS0<2>	IEC0<2>	IPC0<20:18>	IPC0<17:16>
INT0 – External Interrupt 0	3	3	IFS0<3>	IEC0<3>	IPC0<28:26>	IPC0<25:24>
T1 – Timer1	4	4	IFS0<4>	IEC0<4>	IPC1<4:2>	IPC1<1:0>
IC1 – Input Capture 1	5	5	IFS0<5>	IEC0<5>	IPC1<12:10>	IPC1<9:8>
OC1 – Output Compare 1	6	6	IFS0<6>	IEC0<6>	IPC1<20:18>	IPC1<17:16>
INT1 – External Interrupt 1	7	7	IFS0<7>	IEC0<7>	IPC1<28:26>	IPC1<25:24>
T2 – Timer2	8	8	IFS0<8>	IEC0<8>	IPC2<4:2>	IPC2<1:0>
IC2 – Input Capture 2	9	9	IFS0<9>	IEC0<9>	IPC2<12:10>	IPC2<9:8>
OC2 – Output Compare 2	10	10	IFS0<10>	IEC0<10>	IPC2<20:18>	IPC2<17:16>

- Macros are provided in the device header file to simplify/clarify usage. The PIC32MX370F512L header file is located here:
 - `/xc32/<version>/pic32mx/include/proc/p32mx370f512l.h`

```

40654
40655  /* Vector Numbers */
40656  #define _CORE_TIMER_VECTOR
40657  #define _CORE_SOFTWARE_0_VECTOR
40658  #define _CORE_SOFTWARE_1_VECTOR
40659  #define _EXTERNAL_0_VECTOR
40660  #define _TIMER_1_VECTOR
40661  #define _INPUT_CAPTURE_1_VECTOR
40662  #define _OUTPUT_COMPARE_1_VECTOR
40663  #define _EXTERNAL_1_VECTOR
40664  #define _TIMER_2_VECTOR
40665  #define _INPUT_CAPTURE_2_VECTOR
40666  #define _OUTPUT_COMPARE_2_VECTOR

```

IPLx[SRS|SOFT|AUTO]

- Assign Interrupt priority level (0-7) to the ISR, and
- Define how context is preserved
 - The generated code preserves context by either using the shadow register set (SRS) or using generated software instructions (SOFT) to push context onto the stack.

- AUTO setting tells the compiler to use the run-time value in CP0 register SRSCtl to determine whether it should use SRS or SOFT context-saving mode.

isrName

- Unique name you give the ISR.

Example

In this example, we define a Timer 2 ISR function ***T2Interrupt()*** that is configured as a IPL-level-7 process, and to which is assigned the shadow-register set (as defined by the **DEVCFG3**_{FSRSSEL<2:0>} (Device Configuration Word 3) bit settings shown):

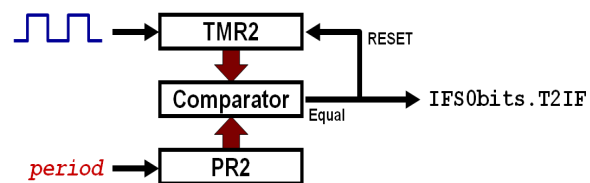
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Step 3. Configure Peripheral

Next, you must configure the peripheral to generate interrupt request events.

For example, the PIC32MX contains several Timer peripherals modules. Each module has a period register (PRx) that, when properly initialized, will periodically trigger a Timer interrupt request signal in an IFSx register as shown:



In this example, we will initialize Timer 2 to generate interrupt requests every 100mS, given a peripheral bus clock (PBCLK) of 10 MHz (code not shown):

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```

Step 4. Configure Interrupt Controller

The next step is to configure the associated Interrupt Controller **Flag**, **Enable** and **Priority** settings for the specific interrupt. In this case, those registers/bits associated with Timer 2 interrupt events:

- **IPC2** Register
 - Contains Timer 2 interrupt priority bits
- **IFS0** Register
 - Contains Timer 2 interrupt request bit
- **IEC0** Register
 - Contains the Timer 2 interrupt enable bit

In the following example, we will configure these registers; setting the initial priority level of the Timer 2 interrupt event to 4, ***matching the IPL setting shown in the ISR function definition below.***

```

1#include <xc.h>
2#include <sys/attribs.h>
3
4/* Config bits (not all shown) */
5
6#pragma config FSRSEL = PRIORITY_7 /* Assign the SRS to level 7 priority handlers */
7
8void __ISR(_TIMER_2_VECTOR, IPL7SRS) T2Interrupt(void)
9{
10 // Toggle LED LD1
11 LATGINV = _LATG_LATG6_MASK;
12
13 // Reset interrupt flag
14 IFS0bits.T2IF = 0;
15}
16
17int main(void)
18{
19 // Initialization
20 T2CONbits.TON = 0; /* turn off Timer 2 */
21 T2CONbits.TCKPS = 7; /* pre-scale = 1:256 (T2CLKIN = 39062.5 Hz) */
22 PR2 = 3906; /* T2 period ~ 100mS */
23 TMR2 = 0; /* clear Timer 2 counter */
24
25 IPC2bits.T2IP = 7; /* Set Timer 2 interrupt priority to 7 */
26 IFS0bits.T2IF = 0; /* Reset Timer 2 interrupt flag */
27 IEC0bits.T2IE = 1; /* Enable interrupts from Timer 2 */
28

```

```

29 while(1);           /* main application loop */
30
31}

```

Step 5. Assign The Shadow Register Set to an IPL

The PIC32MX has **1 shadow register set** that can be used for any priority level, eliminating software context switch and reducing interrupt latency.

If you assign a shadow register set to an interrupt handler function (as in our example), you **must also** initialize the **DEVCFG3_{FSRSSEL<2:0>}** (**Device Configuration Word 3**) bit settings.

The following example initializes DEVCFG3_{FSRSSEL<2:0>} so that the shadow register sets is assigned to interrupt priority level 7.

```

1#include <xc.h>
2#include <sys/attribs.h>
3
4/* Config bits */
5#pragma config FSRSEL = PRIORITY_7 /* Assign the SRS to level 7 priority handlers */
6
7void __ISR (_TIMER_2_VECTOR, IPL7SRS) T2Interrupt(void)
8{
9    // Toggle LED LD1
10   LATGINV = _LATG_LATG6_MASK;
11
12   // Reset interrupt flag
13   IFS0bits.T2IF = 0;
14}
15
16int main(void)
17{
18   // Initialization
19   T2CONbits.TON = 0; /* turn off Timer 2 */
20   T2CONbits.TCKPS = 7; /* pre-scale = 1:256 (T2CLKIN = 39062.5 Hz) */
21   PR2 = 3906; /* T2 period ~ 100ms */
22   TMR2 = 0; /* clear Timer 2 counter */
23
24   IPC2bits.T2IP = 7; /* Set Timer 2 interrupt priority to 7 */
25   IFS0bits.T2IF = 0; /* Reset the Timer 2 interrupt flag */
26   IEC0bits.T2IE = 1; /* Enable interrupts from Timer 2 */
27
28   while(1); /* main application loop */
29
30}

```

Step 6. Set Interrupt Controller Mode

Next, we need to configure the interrupt controller to operate in MULTI-VECTOR MODE, whereby each interrupt source is assigned to its own interrupt vector.

This setting is controlled by the **INTCON_{MVEC}** bit.

MPLAB[®] XC32 provides a nice macro that can be applied with the INTCONSET register as shown in the following example:

```

1 #include <xc.h>
2 #include <sys/attribs.h>
3 /* Config bits */
4 #pragma config FSRSEL = PRIORITY_7    /* Assign the SRS to level 7 priority handlers */
5
6 void __ISR (_TIMER_2_VECTOR, IPL7SRS) T2Interrupt(void)
7 {
8     // Toggle LED LD1
9     LATGINV = _LATG_LATG6_MASK;
10
11     // Reset interrupt flag
12     IFS0bits.T2IF = 0;
13 }
14
15 int main(void)
16 {
17     // Initialization
18     T2CONbits.TON = 0;          /* turn off Timer 2 */
19     T2CONbits.TCKPS = 7;        /* pre-scale = 1:256 (T2CLKIN = 39062.5 Hz) */
20     PR2 = 3906;                 /* T2 period ~ 100ms */
21     TMR2 = 0;                   /* clear Timer 2 counter */
22
23     IPC2bits.T2IP = 7;          /* Set Timer 2 interrupt priority to 7 */
24     IFS0bits.T2IF = 0;          /* Reset the Timer 2 interrupt flag */
25     IEC0bits.T2IE = 1;          /* Enable interrupts from Timer 2 */
26
27     INTCONSET = _INTCON_MVEC_MASK; /* Set the interrupt controller for multi-vector
28 mode */
29
30     while(1);                   /* main application loop */
31
32 }

```

Step 7. Enable Interrupt Exceptions

Next, we need to **globally enable interrupts** by setting the CP0 **Status_{IE}** bit.

MPLAB[®] XC32 provides a handy **__builtin()** function for this task:

```

#include <xc.h>
1#include <sys/attribs.h>
2
3/* Config bits */
4#pragma config FSRSEL = PRIORITY_7    /* Assign the SRS to level 7 priority handlers */
5
6void __ISR (_TIMER_2_VECTOR, IPL7SRS) T2Interrupt(void)
7{
8    // Toggle LED LD1
9    LATGINV = _LATG_LATG6_MASK;
10
11    // Reset interrupt flag
12    IFS0bits.T2IF = 0;
13}
14
15int main(void)
16{
17    // Initialization
18    T2CONbits.TON = 0;          /* turn off Timer 2 */
19    T2CONbits.TCKPS = 7;        /* pre-scale = 1:256 (T2CLKIN = 39062.5 Hz) */
20    PR2 = 3906;                 /* T2 period ~ 100mS */
21    TMR2 = 0;                   /* clear Timer 2 counter */
22
23    IPC2bits.T2IP = 7;          /* Set Timer 2 interrupt priority to 7 */
24    IFS0bits.T2IF = 0;          /* Reset the Timer 2 interrupt flag */
25    IEC0bits.T2IE = 1;          /* Enable interrupts from Timer 2 */
26
27    INTCONSET = _INTCON_MVEC_MASK; /* Set the interrupt controller for multi-vector
28mode */
29
30    __builtin_enable_interrupts(); /* Set the CP0 Status IE bit to turn on interrupts globally */
31
32    while(1);                   /* main application loop */
33
34}

```

Step 8. Enable Peripheral

Finally, we trigger the generation of interrupt exceptions by enabling the peripheral(s):

```

1#include <xc.h>
2#include <sys/attrs.h>
3
4/* Config bits */
5#pragma config FSRSEL = PRIORITY_7    /* Assign the SRS to level 7 priority handlers */
6
7void __ISR (_TIMER_2_VECTOR, IPL7SRS) T2Interrupt(void)
8{
9    // Toggle LED LD1
10    LATGINV = _LATG_LATG6_MASK;
11
12    // Reset interrupt flag
13    IFS0bits.T2IF = 0;
14}
15
16int main(void)
17{
18    // Initialization
19    T2CONbits.TON = 0;          /* turn off Timer 2 */
20    T2CONbits.TCKPS = 7;        /* pre-scale = 1:256 (T2CLKIN = 39062.5 Hz) */
21    PR2 = 3906;                 /* T2 period ~ 100mS */
22    TMR2 = 0;                   /* clear Timer 2 counter */
23
24    IPC2bits.T2IP = 7;          /* Set Timer 2 interrupt priority to 7 */
25    IFS0bits.T2IF = 0;          /* Reset the Timer 2 interrupt flag */
26    IEC0bits.T2IE = 1;          /* Enable interrupts from Timer 2 */
27
28    INTCONSET = _INTCON_MVEC_MASK; /* Set the interrupt controller for multi-vector
29mode */
30
31    __builtin_enable_interrupts(); /* Set the CP0 Status IE bit to turn on interrupts globally */
32
33    T2CONbits.TON = 1;          /* Enable Timer 2 peripheral */
34
35    while(1);                   /* main application loop */
36

```

4. Tasks

Implement “Push-on/Push-off Buttons”. Include non-blocking button press and release detection using change notification interrupts. The requirements are:

Write the code that converts the three momentary contact pushbuttons to operate as push-on/push-off buttons and that satisfies the following specifications:

- The state of the three push buttons BTNR, BTNC, and BTND are displayed on LED0 through LED2.

- The function provides for a 20 ms debounce period for every press and release of a push button.
- The new button state is reported at the start of the 20 ms debounce period.
- The button polling function is to be non-blocking during the 20 ms debounce period.